Approximate Bayesian Inference

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Recapitulation

1. what principle is used in Jeffrey's prior
2. is improper prior $p(\theta) \propto 1$ always non-informative?
3. what is conjugate prior

What is the result of Bayesian inference?

The Bayes rule

$$p(\theta|Y) \propto p(Y|\theta)p(\theta)$$

in practice we appreciate moments, HPD regions:

$$\mathrm{E}(\theta) = \int \theta p(\theta|Y) d\theta \qquad \mathrm{E}(g(\theta)) = \int g(\theta) p(\theta|Y) d\theta$$

$$C : \{\mu \pm 3\sigma\}$$

These are available only for (most of) proper distributions.

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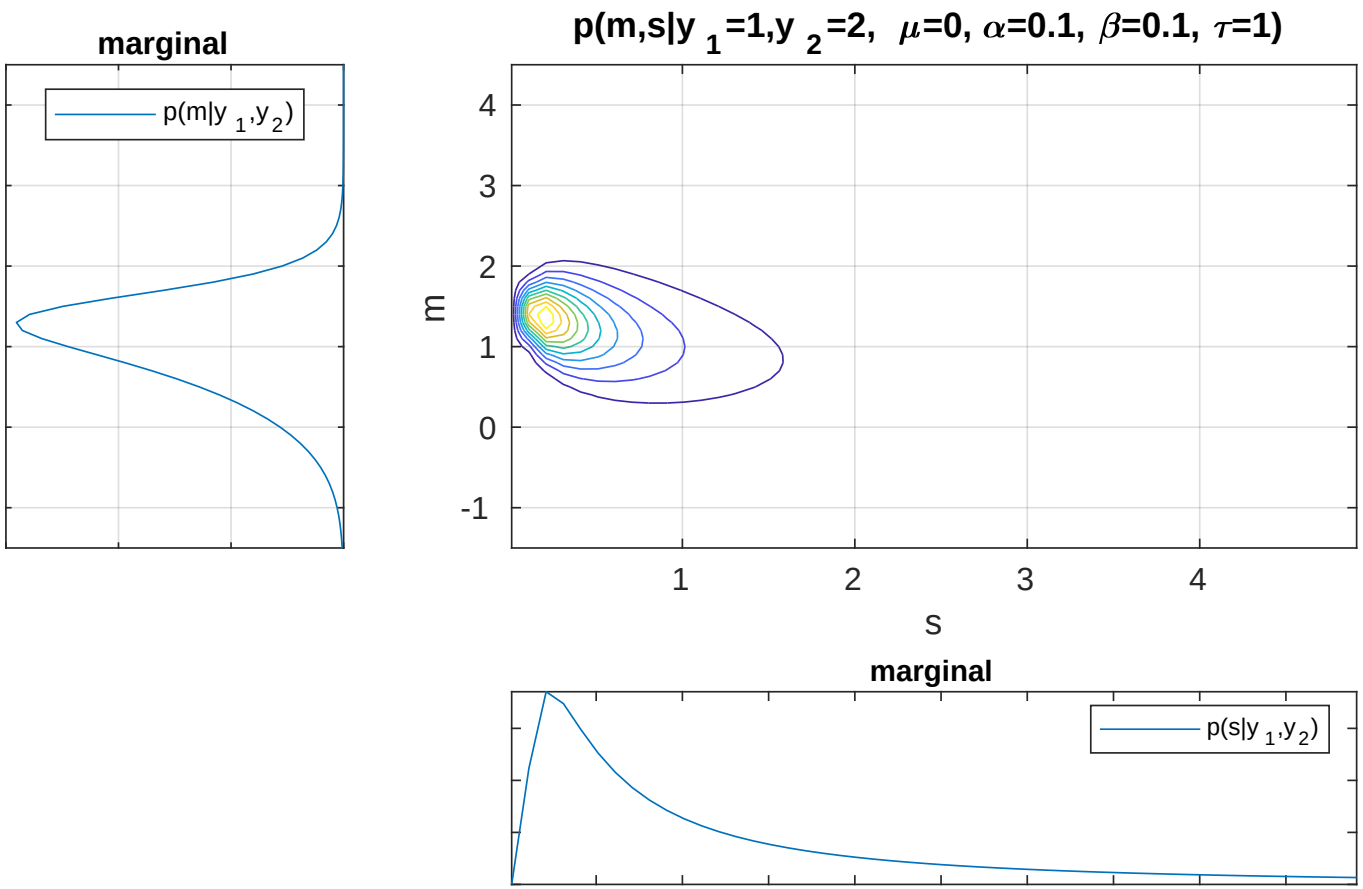
What if we do not have that?

1. choose different metrics
2. approximate $p(\theta|Y) \approx q(\theta|Y)$
3. combine the above

Example


marginal
p(m|y 1,y 2)
p(m,s|y 1=1,y 2=2, μ=0, α=0.1, β=0.1, τ=1)
m
s
4
3
2
1
0
-1
1
2
3
4
marginal
p(s|y 1,y 2)

Alternative results of Bayesian inference

Maximum likelihood (ML): is Bayesian for improper prior $p(\theta) \propto 1$

$$\hat{\theta} = \arg\max p(Y|\theta)$$

Maximum aposteriori (MAP): is a point estimate for prior $p(\theta)$

$$\hat{\theta} = \arg\max p(Y|\theta)p(\theta)$$

Maximum marginal likelihood: is point estimate of a marginal $\theta = [\theta_1, \theta_2]$

$$\hat{\theta}_1 = \arg\max_{\theta_1} p(\theta_1|Y) = \arg\max_{\theta_1} \int p(\theta_1, \theta_2|Y) d\theta_2$$

Empirical Bayes (EB): is a method of inferring $\theta = [\theta_1, \theta_2]$

$$p(\theta_1|Y) \approx p(\theta_1|Y, \theta_2^*)$$

where $\theta_2^* = \arg\max_{\theta_2} p(\theta_2|Y)$.

Distributional approximations

Principles of approximations:

Laplace Approximation: by Taylor approximation

$$\log p(\theta) \approx \log p(\hat{\theta}) + [\nabla \log p(\hat{\theta})]^T(\theta - \hat{\theta}) - \frac{1}{2}(\theta - \hat{\theta})H(\theta - \hat{\theta})$$

$$H = -\nabla\nabla \log p(\hat{\theta})$$

Divergence Minimization: approximation by optimization

Monte Carlo: approximation by sampling

Divergence minimization

We seek best approximation of intractable distribution $p(\theta)$ in the chosen class of parametric functions, $q(\theta|)$, such that

$$\theta^* = \arg\min_{\theta} D(p, q),$$

where $D(p, q)$ is a statistical divergence.

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Variational Bayes:

1. conditional independence $q(\theta_1, \theta_2) = q(\theta_1)q(\theta_2)$,
2. (reverse) Kullback-Leibler. $\mathrm{KL}(q||p) \neq \mathrm{KL}(p||q)$

Kullback Leibler divergence

Measure of divergence between two probability densities

$$KL(q||p) = E_q\left(\log\frac{q}{p}\right)$$

Not a metric!

$$E_q\left(\log\frac{q}{p}\right) \neq E_p\left(\log\frac{p}{q}\right)$$

also known as relative/free entropy

$$KL(q||p) = E_q\left(\log q\right) - E_q\left(\log p\right)$$

with properties:

1. $KL(q||p) \geq 0$,
2. $KL(q||p) = 0, \iff q = p$

EM algorithm is a divergence minimization problem

Approximation of posterior $p(\theta_1, \theta_2|Y)$ under constraints

$$q(\theta_1, \theta_2|Y) = q(\theta_1|Y, S_1)\delta(\theta_2 - \hat{\theta}_2)$$

reaches minimum of the KL

$$S_1, \hat{\theta}_2 = \arg\min_{S_1, \hat{\theta}_2} KL(q||p)$$

at points:

$$q(\theta_1|Y, S_1) \propto p(\theta_1|Y, \hat{\theta}_2)$$
$$\hat{\theta}_2 = \arg\max E_{q(\theta_1)}\left[\log p(Y, \theta_1, \theta_2)\right].$$

EM algorithm is a coordinate descent solution of this optimization.

Variational Bayes

Is a divergence minimization technique with

$$q^* = \arg\min_q KL(q||p) = \arg\min_q E_q\left(\log\frac{q}{p}\right)$$

$$q(\theta_1, \theta_2) = q(\theta_1|Y)q(\theta_2|Y).$$

which allows free-form optimization.

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$$q(\theta_1,\theta_2) = q(\theta_1|Y)q(\theta_2|Y).$$

which allows free-form optimization.
Result:

$$q(\theta_1|Y) \propto \exp\left(E_{q(\theta_2)}\left[\log p(Y,\theta_1,\theta_2)\right]\right)$$
$$q(\theta_2|Y) \propto \exp\left(E_{q(\theta_1)}\left[\log p(Y,\theta_1,\theta_2)\right]\right)$$

which is a set of implicit functions.

- Proportionality above allows to use $p(Y,\theta_1,\theta_2)$ in place of $p(\theta_1,\theta_2|Y)$
- Variational EM algorithm (E-E algorithm).

E-step: m of the toy problem

Proxy distribution

$$q(m) = \mathrm{E}_{q(s)}(\log p(y_1, y_2, m, s))$$

using

$$p(y_1, y_2, m, s) \propto \frac{1}{s}\frac{1}{s^{\alpha_0+1}} \exp\left(-\frac{1}{2}\frac{(m-y_1)^2}{s} - \frac{1}{2}\frac{(m-y_2)^2}{s} - \frac{1}{2}\frac{(m-\mu)^2}{\tau} - \frac{\beta_0}{s}\right)$$

$$\begin{aligned}
\log p(y_1, y_2, m, s) &\propto (\alpha_0 + 2)\log s - \frac{1}{2}\frac{(m-y_1)^2}{s} - \frac{1}{2}\frac{(m-y_2)^2}{s} - \frac{1}{2}\frac{(m-\mu)^2}{\tau} - \frac{\beta_0}{s}, \\
&\propto -\frac{1}{2}\mathrm{E}\left(\frac{1}{s}\right)\left[(m-y_1)^2 + (m-y_2)^2\right] - \frac{1}{2}\frac{(m-\mu)^2}{\tau} \\
&\propto -\frac{1}{2}\left[\frac{(m-y_1)^2}{\hat{s}} + \frac{(m-y_2)^2}{\hat{s}}\right] - \frac{1}{2}\frac{(m-\mu)^2}{\tau}
\end{aligned}$$

$$q(m) = \mathcal{N}\left(m; \left(\frac{1}{\tau} + \frac{2}{\hat{s}}\right)^{-1}\left(\frac{\mu}{\tau} + \frac{y_1 + y_2}{\hat{s}}\right), \left(\frac{1}{\tau} + \frac{2}{\hat{s}}\right)^{-1}\right)$$

E-step: s

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$$\log p(y_1, y_2, m, s) = -(\alpha_0+2)\log s - \frac{1}{2}\frac{(m-y_1)^2}{s} - \frac{1}{2}\frac{(m-y_2)^2}{s} - \frac{1}{2}\frac{(m-\mu)^2}{\tau} - \frac{\beta_0}{s},$$

$$\begin{aligned}
\mathrm{E}_{q(m)}(\log p(y_1, \cdot)) &= -(\alpha_0+2)\log s - \frac{1}{2s}\mathrm{E}_{q(m)}\left[(m-y_1)^2 + (m-y_2)^2\right] - \frac{\beta_0}{s} \\
&= -(\alpha_0+2)\log s - \frac{1}{2s}\mathrm{E}_{q(m)}\left[(m^2 - 2my_1 + y_1^2 + m^2 - 2y_2 m + y_2^2\right]
\end{aligned}$$

$$\begin{aligned}
q(s) &= i\mathcal{G}(\alpha, \beta), \\
\alpha &= \alpha_0 + 1, \\
\beta &= 0.5\mathrm{E}(m^2) - \mathrm{E}(m)(y_1 + y_2) + 0.5(y_1^2 + y_2^2) + \beta_0
\end{aligned}$$

Toy: Variational Bayes

Factors:

$$q(m) = \mathcal{N}\left(m; \left(\frac{1}{\tau} + \frac{2}{\hat{s}}\right)^{-1} \left(\frac{\mu}{\tau} + \frac{y_1 + y_2}{\hat{s}}\right), \left(\frac{1}{\tau} + \frac{2}{\hat{s}}\right)^{-1}\right)$$

$$q(s) = i\mathcal{G}\left(\alpha_0 + 1, \mathrm{E}(m^2) - \mathrm{E}(m)(y_1 + y_2) + 0.5(y_1 + y_2) + \beta_0\right)$$

with

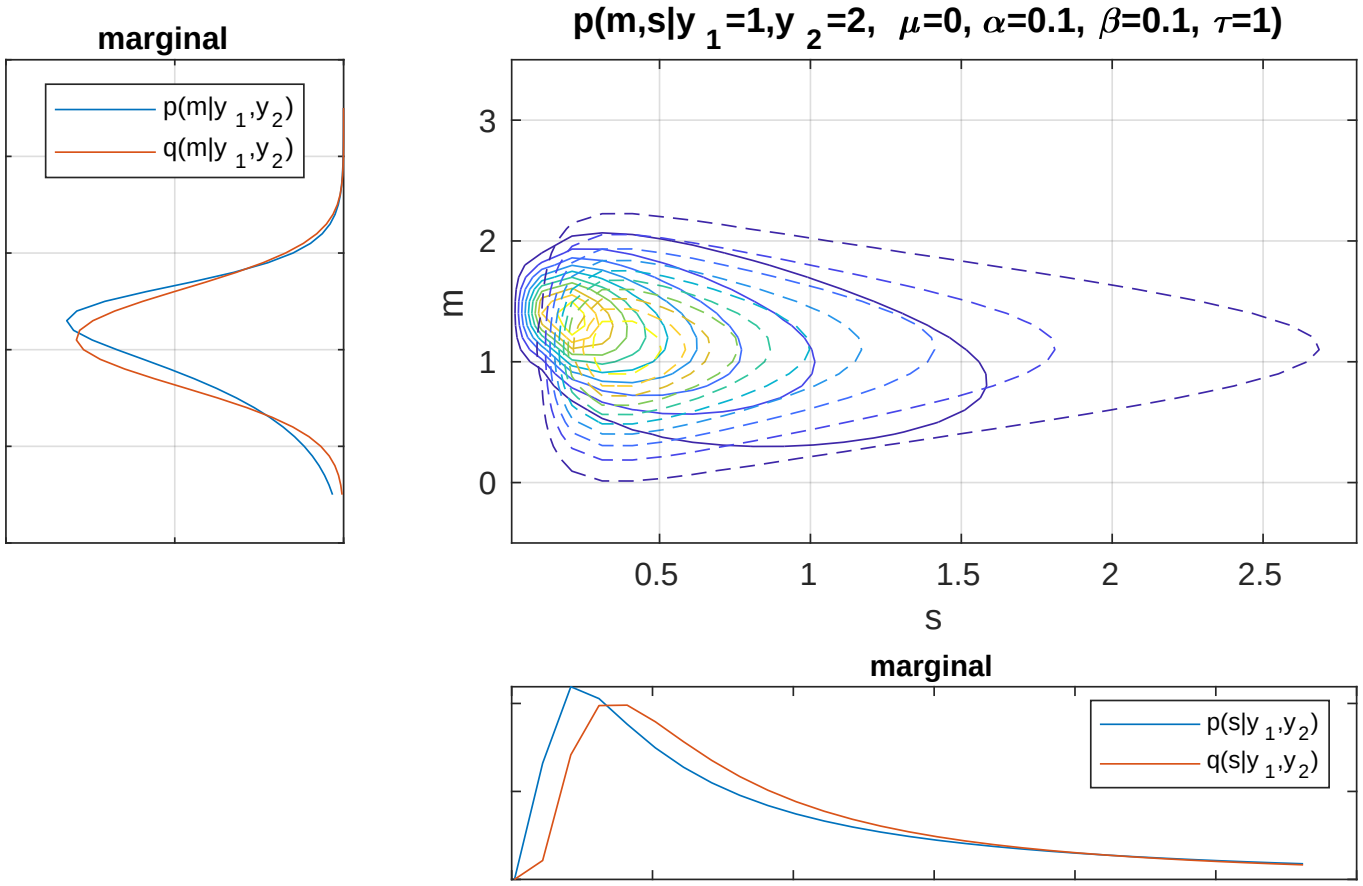
$$\hat{s} = \frac{\mathrm{E}(m^2) - \mathrm{E}(m)(y_1 + y_2) + 0.5(y_1 + y_2) + \beta_0}{\alpha_0 + 1},$$

$$\mathrm{E}(m) = \left(\frac{1}{\tau} + \frac{2}{\hat{s}}\right)^{-1} \left(\frac{\mu}{\tau} + \frac{y_1 + y_2}{\hat{s}}\right),$$

$$\mathrm{E}(m^2) = \mathrm{E}(m)^2 + \left(\frac{1}{\tau} + \frac{2}{\hat{s}}\right)^{-1},$$

which needs to be (Iterated).

Toy: Variational Bayes Iterations



Monte Carlo methods

Approximation of a distribution by "Dirac train"

$$p(x) \approx \frac{1}{N} \sum_{i=1}^{N} \delta(x - x^{(i)}).$$

Approximation of moments, cumulative density.

1. Importance sampling,
   - 1.1 Adaptive importance sapling
   - 1.2 Population Monte Carlo
2. Monte Carlo Markov Chain
   - 2.1 Metropolis-Hastings (Gibbs sampler)
   - 2.2 Hybrid MC (Hamiltonian Monte Carlo)

Convergence assured under mild conditions, different convergence rate.

Importance Sampling

To represent

$$p(x) \approx \frac{1}{N} \sum_{i=1}^{N} \delta(x - x^{(i)}). \tag{1}$$

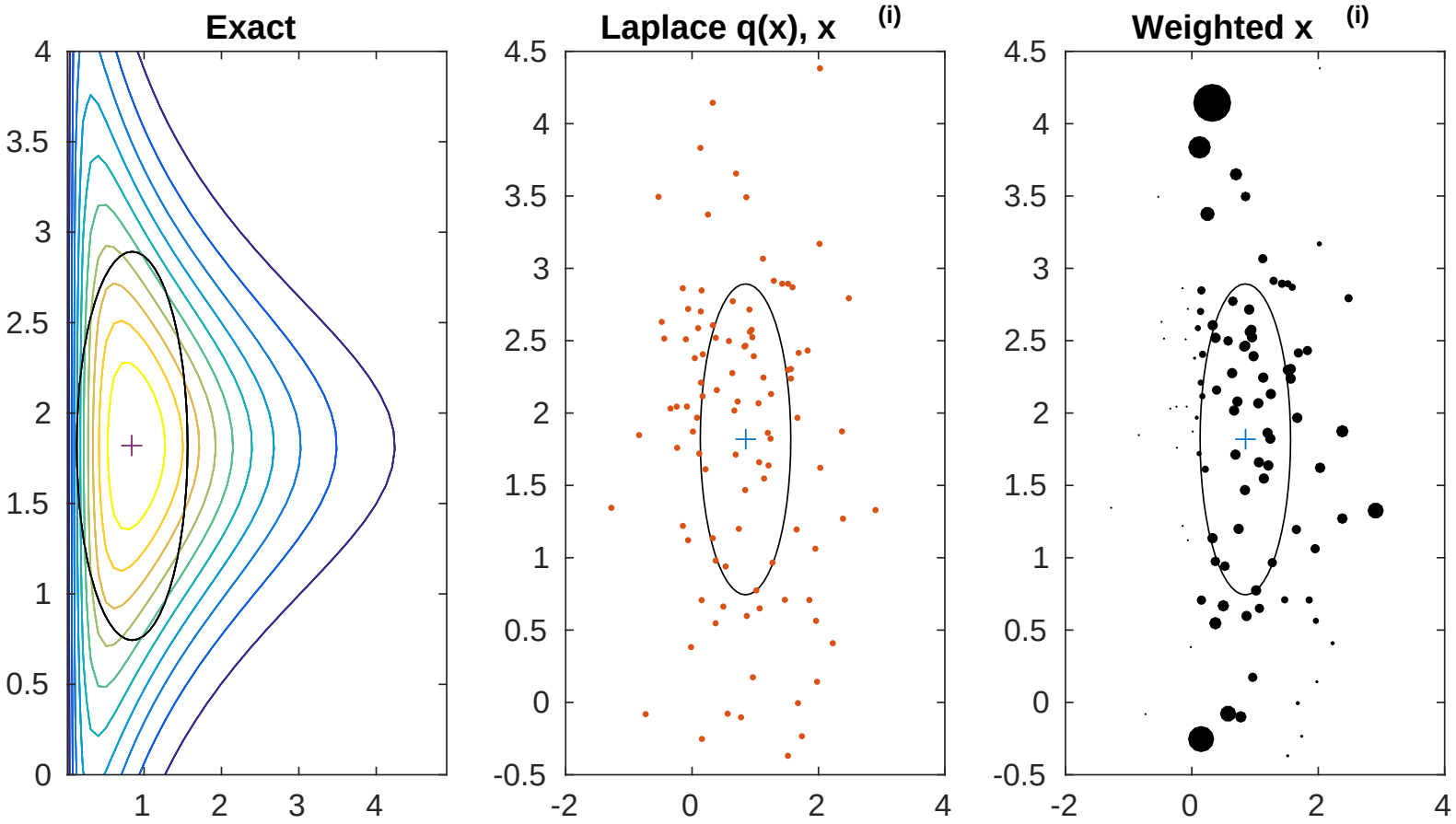
an ideal sampler should sample $x^{(i)} \sim p(x)$, which is not available. Using

$$p(x) = p(x) \frac{q(x)}{q(x)},$$

we can approximate $q(x)$ by (1) by sampling $x^{(i)} \sim q(x)$.

$$\begin{aligned}
p(x) &\propto \frac{p(x)}{q(x)} \frac{1}{N} \sum_{i=1}^{N} \delta(x - x^{(i)}), \\
&\propto \sum_{i=1}^{N} \tilde{w}_i \delta(x - x^{(i)}), && \tilde{w}_i = \frac{p(x^{(i)})}{q(x^{(i)})} \\
&= \sum_{i=1}^{N} w_i \delta(x - x^{(i)}) && w_i = \frac{\tilde{w}_i}{\sum_{i=1}^{N} \tilde{w}_i}
\end{aligned}$$

Toy: Importance sampling



MCMC: Metropolis Hastings

Instead of fixed distribution, we define a Markov chain that converges to the true distribution.

1. choose transition kernel $q(x|x^{(i)})$,
2. generate sample $x^* \sim q(x|x^{(i)})$,
3. With probability
$$\min\left(1, \frac{p(x^*)q(x^{(i)}|x^*)}{p(x^{(i)})q(x^*|x^{(i)})}\right)$$
accept $(i = i + 1, x^{(i)} = x^*)$, else reject; goto 2.

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How to choose the kernel:

- Random walk (Gaussian), with parameters θ
- Use known properties: conditionals

MCMC: Gibbs sampler

Special case of MH for mutidimensional distributions.

$$p(x_1, x_2, \ldots, x_k)$$

with MH probability of acceptance equal to one.

1. generate sample $x_1^{(i+1)} \sim p(x_1 | x_2^{(i)}, \ldots x_k^{(i)})$,
2. generate sample $x_2^{(i+1)} \sim p(x_2 | x_1^{(i+1)}, \ldots x_k^{(i)})$,

   $\vdots$

3. generate sample $x_k^{(i+1)} \sim p(x_2 | x_1^{(i+1)}, \ldots x_{k-1}^{(i+1)})$,

Suitable when these distributions are tractable.

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Suitable when these distributions are tractable.

- not suitable for parallel computing

Toy: Gibbs sampler

Joint

$$p(y_1, y_2, m, s) \propto \frac{1}{s} \frac{1}{s^{\alpha_0+1}} \exp\left(-\frac{1}{2}\frac{(m-y_1)^2}{s} - \frac{1}{2}\frac{(m-y_2)^2}{s} - \frac{1}{2}\frac{(m-\mu)^2}{\tau} - \frac{\beta_0}{s}\right)$$

Conditionals:

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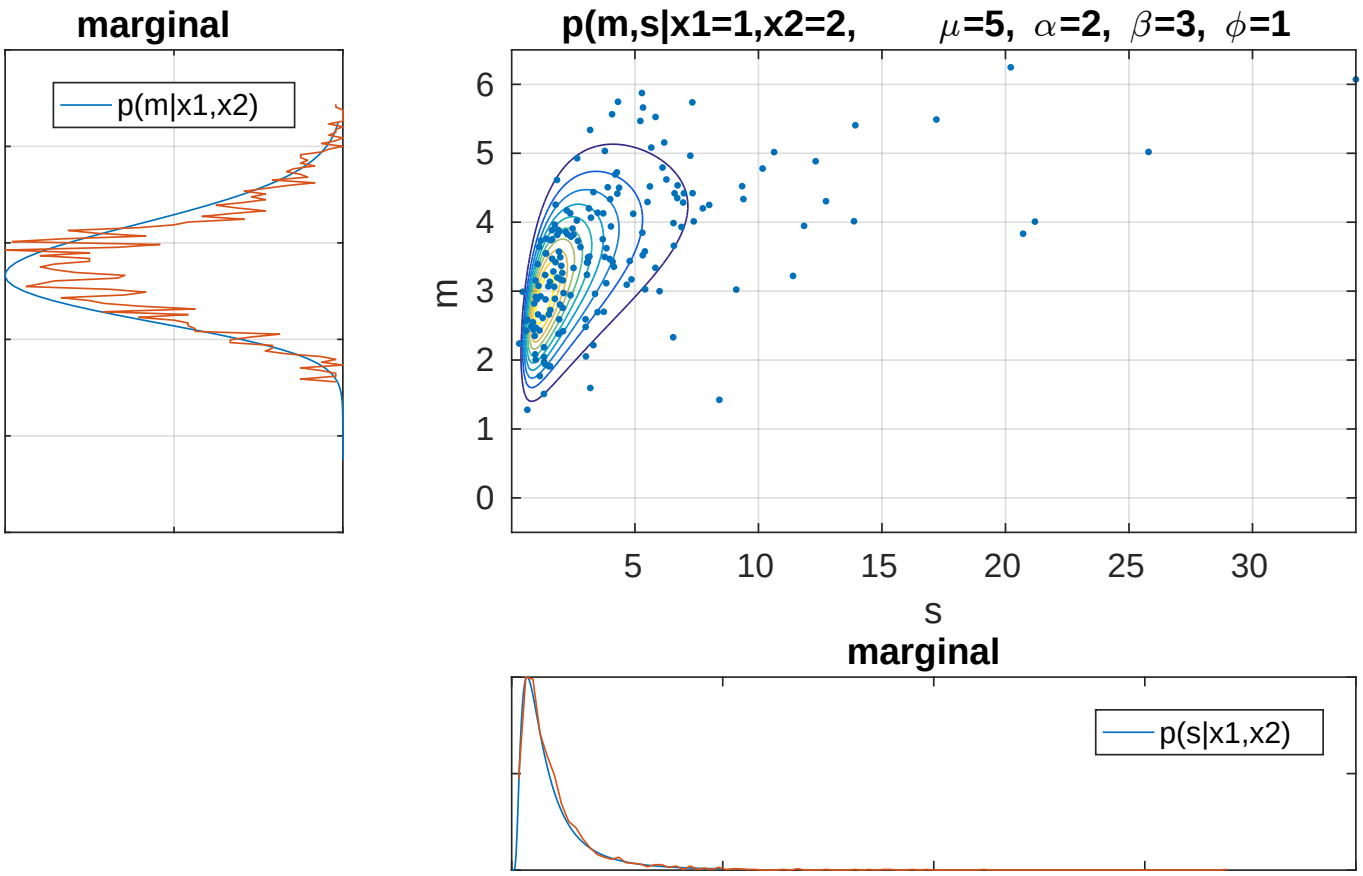
Toy: Gibbs sampler



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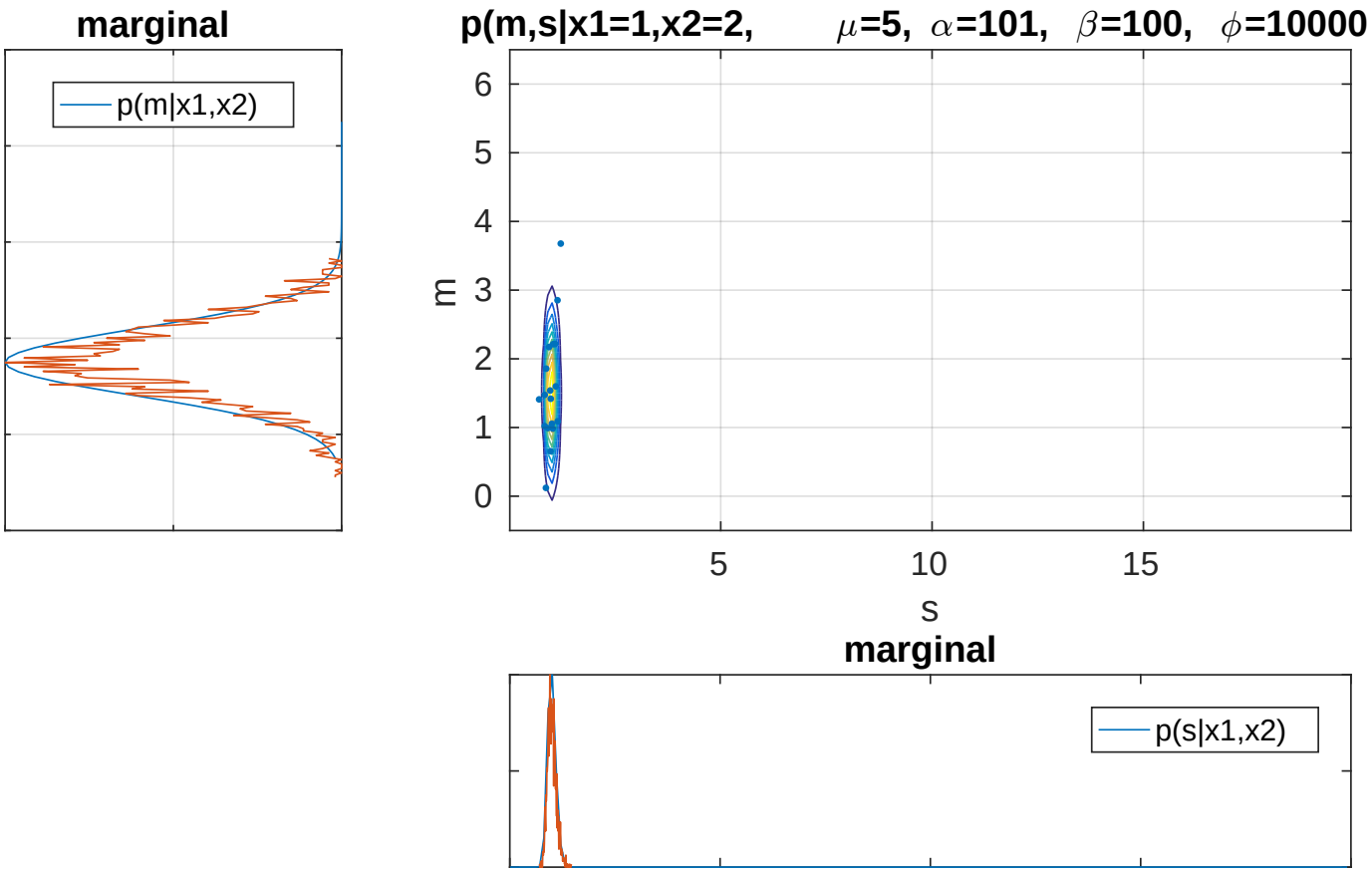

Homework assignment

1. Working code for Variational Bayes estimation of the toy problem (3 points).

$$p(m, s|\tau, Y)$$

2. Working code for Gibbs sampler of the toy problem (3 points)

$$p(m, s|\tau, Y)$$

3. Any of them for known s unknown τ

$$p(m, \tau|s, Y)$$